WildTrace: Photo-to-Pen Robot Drawing with Learned Path-Following Policies and a Calibrated Digital Twin

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***Abstract*—We present WildTrace, an end-to-end system that turns photographs of animals and faces into pen-stroke trajectories and draws them with a 7-DoF UFACTORY xArm 7, both in a calibrated NVIDIA Isaac Sim twin and on the physical arm. A medallion data pipeline uses a diffusion model to redraw each photo as a clean outline, an OpenCV prescreen with retries to reject bad outlines, and a vectoriser that emits a few ordered strokes, yielding 2,874 validated trajectories over six animal classes. On top of this data we train an Action Chunking Transformer (ACT) that maps joint state and a short window of the upcoming path to joint steps, using only kinematic inverse-kinematics labels with noise injection. On a held-out set of 48 drawings in the twin at 1 g, ACT finishes 48/48 drawings against 45/48 for a Lula IK baseline, lowers mean path RMSE from 0.277 mm to 0.192 mm (lower on all 45 paired drawings, Wilcoxon $p < 10^{-12}$), a sixth of a marker line, halves joint jerk, and is unaffected by $\pm$12 mm paper height and 21 mm planar placement errors. IK tuned over eight settings finishes all 48 at 0.262 mm and keeps a lower worst-point error, but ACT is lower on every drawing; a streaming IK without the settle gate gains only 0.010 mm and gravity-compensated IK 0.011 mm, so the improvement comes from the learned policy; without its noise-injected training episodes the same policy fails every drawing. A fine-tuned SmolVLA re-planned every tick matches ACT within its seed spread (0.180 mm) but is jerkier and twice as slow to query, and a rule-based supervisor that hands stalls and ink gaps to IK repairs injected ink skips on all 48 drawings (232 redraws for 345 skips, none abandoned) at 0.001 mm cost. On the physical arm, ACT completes 30/30 animal drawings at 0.532 mm RMSE and three runs of two face portraits at 0.497 mm, about 2.7$\times$ the twin, a gap that we analyse.**

***Index Terms*—robot drawing, imitation learning, action chunking, vision-language-action models, digital twin, sim-to-real**

I. Introduction

Robot drawing is a compact testbed for precise contact-rich path following: the pen must stay on a sheet a fraction of a millimetre away, follow tight curves without overshoot, and lift and land cleanly between strokes. Every error is visible on paper. Classic pipelines solve inverse kinematics (IK) per waypoint and wait for the arm to settle [1], [2], which is reliable but slow and leaves a visible tremor. Learned visuomotor policies such as ACT [3], diffusion policy [4] and vision-language-action (VLA) models such as SmolVLA [5] promise smoother motion, but are usually evaluated by task success rather than by sub-millimetre path accuracy on open-ended content.

WildTrace closes the loop from an arbitrary photograph to ink on paper (Fig. 1) and uses it to compare controllers on the same drawings under identical physics. Our contributions are:

1) A photo-to-trajectory pipeline (diffusion redraw, prescreen and retry, vectoriser) that builds 2,874 validated drawing trajectories and a 48-drawing held-out test set, with a benchmark of four generators and four open-weight VLM judges (Sec. III).
2) A path-conditioned ACT policy trained only on kinematic IK labels with noise injection and a capped progress projection, which beats IK in a calibrated 1 g twin on mean accuracy and joint jerk, also after tuning IK, is insensitive to paper placement, and keeps its lead over a streaming IK baseline that shares its progress rule and pace (Sec. IV, V).
3) A like-for-like comparison with a fine-tuned SmolVLA on the same data: re-planned every tick in the batch runner it matches ACT within seed variance, at higher jerk and latency, while the ranking reverses in a second runner, and a negative result on camera-conditioned gap repair.
4) A hybrid supervisor that repairs skipped ink using the run's own bookkeeping, and a physical-arm evaluation on 30 animal drawings and two portraits that quantifies the sim-to-real gap.

The individual methods are established; the contribution is the end-to-end system and a controlled analysis, with ablations, of why a learned path follower beats the IK controller that labelled it.

II. Related Work

Robot drawing and painting. Portrait-drawing robots date back to humanoid systems that extract contours and replay them with IK [1] and to Paul, which emulates an artist's sketching process [2]. Recent systems differ mainly in what they draw and how fast. RoboCoDraw turns a face into a GAN avatar and orders its strokes with a genetic algorithm, drawing in 43 s on a UR5 [9]; AiSketcher draws synthesised portraits with a 3-DoF arm in about two minutes [10]; Nasrat et al. draw portraits in 80 s with line width set by pen height [11]; Piktor-O-Bot adds force-controlled shading on a UR5 and shapes speed by path curvature [12], [13]; TSP-Bot draws



Fig. 1. WildTrace data pipeline on one Open Images [6] photo. Bronze: fetch the image, mask and license, then curate and label (BioCLIP [7]). Silver: isolate the subject and redraw it as an outline with FLUX [8], judged and retried. Gold: the accepted outline becomes a few ordered pen strokes on a normalised canvas (numbered markers show stroke starts).

single-path stipple art with TRAC-IK [14]; and line art can be parsed into stroke graphs for a desktop arm [15]. These systems execute planned paths and report drawing time, style or viewer preference rather than tracking accuracy. Learned stroke models [16], [17] plan in image space, and FRIDA [18] and Spline-FRIDA [19] learn stroke appearance to shrink the sim-to-real gap of the painted image. Closest to our control question, Lee et al. learn motor control with hierarchical reinforcement learning instead of IK, but need pen-tip feedback to correct the learned controller on a real UR5e [20]. We take the strokes as given and measure how precisely learned policies follow them against IK, at sub-millimetre resolution.

Imitation learning for fine control. ACT [3] predicts chunks of future actions with a transformer; diffusion policy [4] models multimodal action distributions. DART [21] injects noise during demonstration to cover off-path states and DAgger [22] aggregates on-policy corrections. We use DART-style noise with a synthetic IK teacher, and DAgger rounds for camera policies.

Vision-language-action models. SmolVLA [5] reuses a pretrained vision-language backbone for control, with a flow-matching action head [23]. We fine-tune SmolVLA on the same data as ACT to test whether a pretrained generalist helps a precision task.

Precise path tracking. Iterative learning control refines a feedforward command over repeated runs of the same path [24], and model-predictive contouring control trades contour error against progress along the path [25]. Both suit a fixed path drawn many times or a known dynamics model; our policies draw each new path once, with no per-path iteration or online optimisation.

Sim-to-real. Domain randomisation [26] widens the training distribution to cover the real system; we instead calibrate the twin to the lab arm and randomise only paper placement.

III. Data Pipeline: Photo to Pen Strokes

The pipeline is a local-first medallion design, in which raw (bronze), cleaned (silver) and final (gold) data are stored as separate layers, with one script per stage and append-only NDJSON manifests, so a stage reruns only for rows whose configuration or input hash changed.

Bronze. Images, segmentation masks and licenses are fetched from Open Images V7 for six classes (bird, cat, dog, fish, frog, horse), de-duplicated, filtered for quality and viewing angle, and labelled with BioCLIP.

Silver. The subject is cropped by its mask and redrawn as a black-on-white outline by FLUX.1-schnell. A LangGraph [27] loop validates each outline with an OpenCV prescreen (foreground ratio, component count, contour size, fragmentation) followed by a local VLM judge (Qwen2.5-VL-7B [28]), and retries with a new seed on failure.

Gold. The best outline per pose is skeletonised and vectorised into a few ordered strokes, resampled and normalised to a unit canvas. The result is 2,874 validated trajectories (bird 370, cat 644, dog 759, fish 166, frog 529, horse 406); 68% were accepted on the first attempt and 2.8% on the fifth, with 1.57 attempts on average. A stratified 48-drawing gold test set (9 bird, 12 cat, 13 dog, 4 fish, 5 frog, 5 horse; mean 4.6 strokes, range 2 to 8, mean 370 points) is never used for training. Faces follow a separate CPU path: MediaPipe [29] face landmarks and hair/body segmentation are traced into a pen-sketch portrait. Fig. 2 shows five portraits from this path next to their source photographs.

Choosing the generator and the judge. Table I compares informative drawings, SD1.5 with ControlNet, FLUX and OmniGen2 [30] on the same 20 subjects, using corrected contour conditioning. Each uses Qwen3-VL-8B in the retry loop with a three-attempt budget. Four judges then rescore the final images, including prescreen rejects, against Qwen3-VL-235B [31]. The OpenCV prescreen favours FLUX's clean line style: informative drawings failed it on all 60 attempts (component count, small contours or lower fragments), so none reached the in-loop judge. This selection effect must be considered alongside the generator scores. The production dataset was judged by Qwen2.5-VL-7B, which agrees poorly with the referee (it passes 75 of 80 final images); in a 186-subject pilot all 54 rejections came from the OpenCV prescreen, so the prescreen and retries, not the VLM, filtered the dataset. Qwen3-VL-8B is the stronger in-loop judge at the same size.

Why open-weight models, not frontier APIs. The generators and judges are open-weight models; we did not evaluate



Fig. 2. Face portraits: Open Images source photographs (top) and the pen-sketch line art traced from MediaPipe landmarks and segmentation, including ears where they stand out from the head (bottom), 15 to 19 strokes each. Photographs CC BY 2.0 via Flickr, by Byron Gonzalez, MangakaMaiden Photography, Luca Boldrini, JJP Bolger Photography and Walt Stoneburner (left to right).

TABLE I
GENERATOR AND JUDGE STUDY (PAPER-R3, 20 SUBJECTS PER GENERATOR). n: EVALUATED IMAGES OR PAIRED VERDICTS. PASS: QWEN3-VL-235B; EXPORT: TRAJECTORY CONSTRAINTS.

Generator	n	Pass	BioCLIP	Blank	Export
Informative drawings	20	0/20	25%	0%	100%
SD1.5 + ControlNet	20	2/20	25%	0%	100%
FLUX.1-schnell	20	16/20	55%	0%	100%
OmniGen2	20	8/20	25%	45%	55%

Judge	n	Agreement	Cohen's κ
Qwen2.5-VL-7B (local)	80	38.8%	0.06
Qwen3-VL-8B	80	88.7%	0.76
Qwen3-VL-30B-A3B	80	91.2%	0.81
Gemma-3-12B	80	78.7%	0.59

proprietary frontier models in either role. The generators and local 7B judge run on the lab's 16 GB laptop GPU; the other judges and the 235B referee are hosted for this study. Open weights support three practical goals. (i) *Volume and cost*: the dataset requires thousands of generate-judge calls; local execution avoids per-call API fees, while the benchmark records hosted costs. (ii) *Privacy*: Studio's face-landmark and segmentation path processes uploaded portraits locally. (iii) *Reproducibility*: pinned checkpoints and saved prompts permit controlled reruns and inspection. Table I measures the agreement of smaller judges with the larger open referee; this is an imperfect reference, not a human evaluation or evidence about proprietary frontier models.

IV. ROBOT, TWIN AND CONTROLLERS

Hardware and twin. A UFACTORY xArm 7 holds a pen in its parallel gripper over a US Letter sheet. The Isaac Sim [32] twin (Fig. 3a) uses the arm's calibrated URDF, the measured



(a) Lab-arm twin (b) Physical xArm 7

Fig. 3. (a) Isaac Sim twin built from the lab arm's calibrated URDF, gripper and measured paper placement, at 1 g. (b) The physical xArm 7 drawing a WildTrace bird.

paper pose and pen touch height, Earth gravity, and a stiff PhysX position servo (stiffness 7.5×10^5, damping 2050) standing in for the vendor joint controller. Two RealSense D435i cameras (wrist and overhead) were located by depth plane fitting and planar PnP on the sheet corners, whose heights above the paper agree to within 1.6%. Each drawing runs a fixed phase machine: approach, lower, draw, lift for every stroke, with short pen-up hops between nearby strokes.

Metrics. Let $\{p_i\}_{i=1}^{N}$ be the planned pen-down samples and $\{e_j\}$ the executed pen-down pen-tip positions, both in 3-D. The nearest-path error of a planned sample is

$$d_i = \min_j \|p_i - e_j\|_2, \tag{1}$$

and we report the path RMSE $\left(\frac{1}{N}\sum_i d_i^2\right)^{1/2}$, the 95th percentile and the maximum of d_i. Because (1) only penalises missing a planned point, we also report the reverse direction, from each executed sample to the plan, which catches stray ink. Joint jerk is the RMS over pen-down samples of $\|\Delta^3 q\|/\Delta t^3$, the third difference of joint positions within a stroke. Corner RMSE is the tracking error at waypoints flagged as corners, and heading error compares the direction of plan and pen over a 4 mm chord.

IK baseline. NVIDIA Lula IK solves joint targets for each waypoint of the resampled path (at most 5 mm apart, densified at corners); joint commands are slew-limited, and the run advances only when the measured tip is within tolerance and its velocity has settled. A waypoint timeout fails the drawing safely. This is the controller the system shipped with and the teacher that labels the policy data.

Streaming IK baseline. To separate learning from the settle gate, a second IK baseline uses exactly ACT's progress rule (the first waypoint ahead of the pen's projection, Sec. IV) with no settle gate, and solves Lula IK every tick for a carrot a fixed arc length L ahead of the pen (pure pursuit). The carrot advances along the path no faster than ACT's median pen speed, 12.4 mm s^{-1}, so both controllers draw at the same pace; we also run it at plain IK's median pen speed, 6.9 mm s^{-1}. Both runs use $L = 1$ mm. The speed cap prevents the carrot from advancing faster than the comparison pace.

ACT path follower. At control tick t the policy observes the robot state $s_t = [\, q_t,\ \dot{q}_t,\ u_{t-1} \,] \in \mathbb{R}^{21}$ (joint positions, velocities and previous joint command) and a path context

$$c_t = \left[\, x_t,\ g_t - x_t,\ \{w_{t,k} - x_t\}_{k=1}^{10},\ \pi_t \,\right] \in \mathbb{R}^{37}, \qquad (2)$$

where x_t is the pen tip, g_t the current target, $w_{t,k}$ ten points spaced at equal arc length along the next 2 cm of path, and π_t the pen state. It predicts a chunk of $H = 10$ joint *increments* $\hat{\delta}_{t:t+H-1}$, trained with an L1 loss against the teacher's increments, and only the first is executed:

$$u_t = u_{t-1} + \alpha\, \hat{\delta}_t, \qquad \alpha = 0.5. \qquad (3)$$

Absolute joint targets were mm-imprecise even when overfitting one drawing, so the policy predicts increments. There is no image input. We use ACT [3] via LeRobot [33] with $d = 256$, 4 encoder and 1 decoder layers, no CVAE, no dropout, learning rate 3×10^{-4}, batch 256 and 40k steps. On a single-drawing overfit gate these choices cut first-action error from 33-45% to 11% of the label, a 10-step chunk fit 2.5× better at p95 than a 30-step chunk, and executing only the first action of each chunk with re-planning every tick tracked better than temporal ensembling. At run time the policy is queried at 60 Hz and half of each predicted step is applied.

Labels without physics. Labels come from Lula IK walked through the same phase machine with no physics simulation (about 20k solves/s), three episodes per training drawing. Two of the three add DART-style command noise [21] and a measured-versus-commanded joint offset, so the policy sees sagged, off-path states with the corrective label. Path progress is the projection of the tip onto the path, capped at 1.5 mm ahead; without the cap progress jumped across narrow hairpins (8.1 mm worst-point error on one frog toe) and spike tips were rounded. The lab-arm model is trained with ±15 mm paper-height jitter. Fine-tuning on 1 g Isaac IK rollouts made tracking worse (0.74 to 0.91 mm on four drawings), because those labels carry IK's own sag (that scene's servo was softer). Removing the noise and offset from the labels, with everything else unchanged, makes the policy fail every test drawing (Sec. V).

Algorithm 1 Hybrid supervisor, run every control tick

Require: policy π, path progress s, inked set $\mathcal{I}$, last advance time t_a

if $t - t_a > 1.5$ s **then**
  drive the next 5 mm of path with Lula IK; $t_a \leftarrow t$ {stall}
else if a stretch $[s_0, s_1] \subset [0, s]$ with $s_1 - s_0 \geq 1.5$ mm has no ink within 1 mm **then**
  hold s; IK goes back to s_0, retraces to s {gap}
  add the retraced path to $\mathcal{I}$; give up after a timeout
else
  $u_t \leftarrow u_{t-1} + \alpha\, \pi(s_t, c_t)_1$ {policy, Eq. (3)}
end if
update s by projecting the tip onto the path (capped at 1.5 mm ahead)

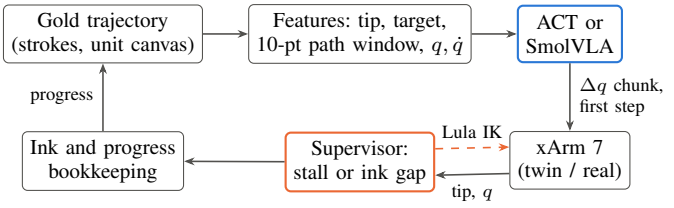


Fig. 4. Control loop. The policy runs every 60 Hz tick on a path window around the current progress; the supervisor hands control to Lula IK only on a stall or an uninked gap (dashed), then hands back.

SmolVLA. We fine-tune SmolVLA [5] on the same dataset (19.8M frames), feeding the full 58-value state through a direct-state MLP, 8 VLM layers, 256 px rendered path images and 20-action chunks (51M of 323M parameters trained). It is served with zero initial noise (better than averaging samples offline); with 3 flow-matching steps, a captured CUDA graph cuts sampling from about 225 ms to 30 ms per chunk.

Hybrid supervisor. A rule-based supervisor (Algorithm 1, Fig. 4) watches the run's own ink bookkeeping rather than the policy. If the drawing target has not advanced for 1.5 s, IK drives the next 5 mm and hands back. If at least 1.5 mm of path behind the pen has no ink within 1 mm (the labeller's coverage rule), IK goes back past the gap and forward again while the run's progress is held, then the policy resumes. Both rules need no camera, so they also work on the physical arm.

V. Results

All simulated runs use the lab-arm twin at 1 g. The main comparison (Table II, Fig. 5) runs all four controllers on all 48 gold drawings with the same code, scene and batch runner; per-run metrics and plots are kept with the code.

ACT against IK. ACT finishes every drawing; IK fails three on waypoint timeouts. On the 45 drawings both finish, ACT has lower path RMSE on all 45: the paired reduction is 0.085 mm (95% CI 0.079 to 0.091), or 31%, with two-sided Wilcoxon signed-rank $p = 5.7 \times 10^{-14}$. The advantage holds in every class (Table III). It is not bought with stray ink: the reverse, pen-to-plan error is also lower (0.374 against 0.397 mm). Over pooled samples (Fig. 5b) the median error

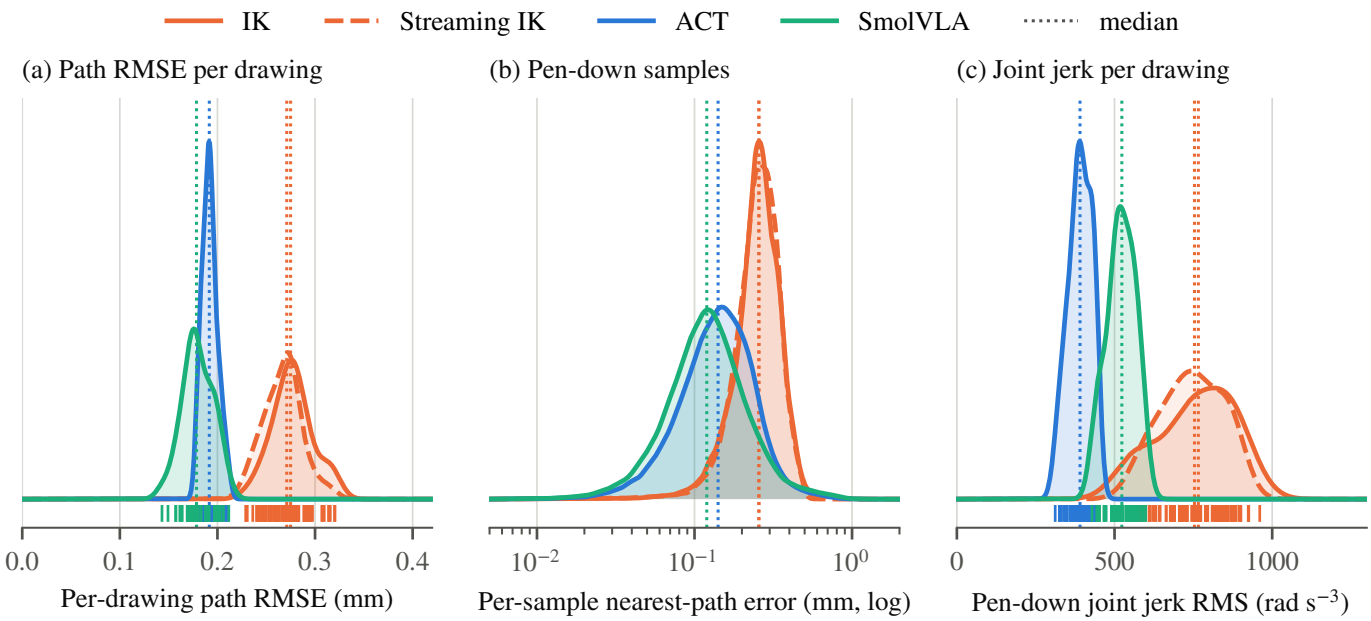


Fig. 5. Kernel density estimates (Gaussian kernel, Scott bandwidth) over the gold set for the four controllers; dotted lines are medians and ticks are individual drawings (n = 45 / 48 / 48 / 48 drawings, 71k / 76k / 76k / 76k pen-down samples for IK / streaming IK / ACT / SmolVLA). (a) Per-drawing path RMSE: the two learned policies separate cleanly from both IK variants. (b) Nearest-path error of every planned pen-down sample (densified plan): medians 0.256, 0.257, 0.141 and 0.120 mm. (c) Pen-down joint jerk: ACT has the lowest, SmolVLA between ACT and IK.

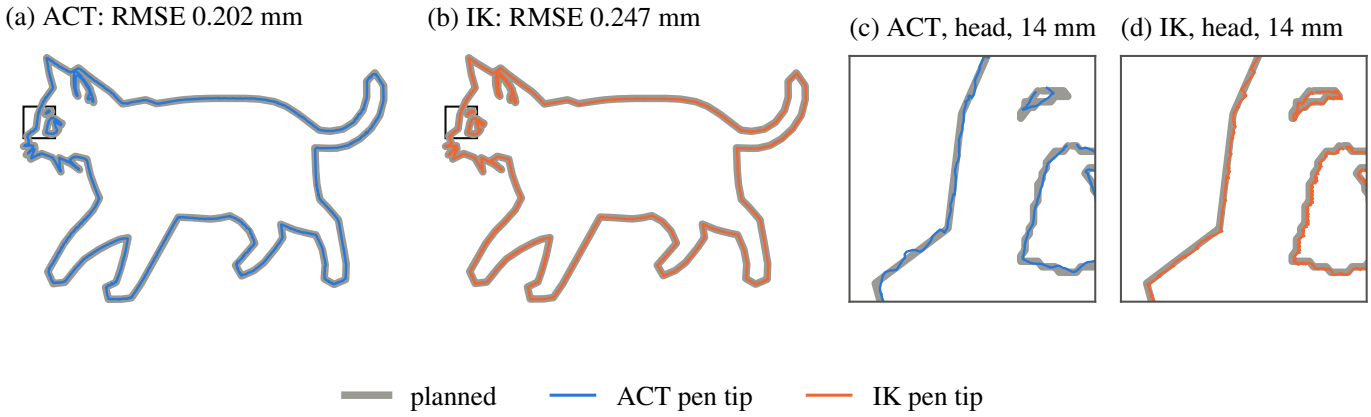


Fig. 6. Planned path (grey) against the executed pen tip for the same gold cat, ACT (blue) and IK (orange), top-down. (c, d) A fixed 14 mm window on the head: at drawing scale both follow the plan closely; the 0.08 mm RMSE difference is a sixth of a marker line and not visible.

falls from 0.256 to 0.141 mm. ACT halves joint jerk (Fig. 5c). Its pen moves at a median 12.4 against 6.9 mm s^{-1}, and IK's pen is settling on 51% of pen-down samples against 18% for ACT. Total draw time is similar (77 against 73 s simulated) because the phase machine's approach, lower and lift moves are shared. The policy server answers in 9.1 ms median, inside the 16.7 ms tick. Retraining ACT with two more seeds gives 0.201 and 0.218 mm on the same 48 drawings (three seeds: 0.204 ± 0.013 mm), all well below both IK variants.

Is the IK baseline tuned? We swept IK's reach tolerance (0.6, 1.2, 2.0 mm), its settle gate (4, 7, 12 mm s^{-1}) and its waypoint step (0.75, 1.5 mm) on the 16 robustness drawings, then ran the best setting, 0.6 mm tolerance with a 12 mm s^{-1} gate, on all 48 (IK* in Table II). It finishes every drawing, is the fastest controller (70.1 s) and has the smallest worst point (0.466 against 0.776 mm for ACT, smaller on all 48 drawings), but its path RMSE improves only to 0.262 mm: ACT is lower on all 48 (paired reduction 0.070 mm, CI 0.064 to 0.076, $p = 7 \times 10^{-15}$) with half the joint jerk. Tightening the gate to 4 mm s^{-1} lost six of 16 drawings to timeouts without lowering the error. ACT's advantage is therefore in mean tracking and joint smoothness, not in its worst point: a third of ACT's worst points (16 of 48) lie within 3 mm of a stroke's start or end, which hold 12% of the path, against 3 of 48 for IK*, so its largest errors come where the pen lands and lifts.

Is it the learning or the settle gate? Streaming IK uses

TABLE II

GOLD TEST SET, 48 HELD-OUT DRAWINGS, LAB-ARM TWIN AT 1 G, SAME CODE AND RUNNER FOR ALL FOUR CONTROLLERS. MEANS OVER FINISHED DRAWINGS; 95% BOOTSTRAP INTERVALS OVER DRAWINGS IN BRACKETS. SEGMENT RMSE MEASURES EACH POINT OF THE DENSIFIED PLAN (FIG. 5B) TO THE EXECUTED POLYLINE INSTEAD OF THE NEAREST SAMPLE, REMOVING SAMPLING DENSITY; SAMPLE-BASED ON THE SAME PLAN IT IS 0.275, 0.268, 0.270, 0.180 AND 0.171 MM. ACT IS LOWER THAN IK ON ALL 45 DRAWINGS BY BOTH. IK* IS LULA IK WITH THE BEST OF EIGHT TOLERANCE, GATE AND STEP SETTINGS; S-IK IS THE STREAMING IK BASELINE AT ACT'S PACE.

Metric	IK	IK*	S-IK	ACT	SmolVLA
Finished	45/48	**48/48**	**48/48**	**48/48**	**48/48**
Path RMSE (mm)	0.277	0.262	0.267	0.192	**0.180**
95% CI	.271-.283	.256-.268	.262-.272	.190-.194	.176-.184
Path p95 (mm)	0.398	0.377	0.373	0.335	**0.327**
Path max (mm)	0.778	**0.466**	0.953	0.776	0.871
Segment RMSE (mm)	0.259	0.255	0.257	0.163	**0.156**
Reverse RMSE (mm)	0.397	0.406	0.401	0.374	**0.356**
Corner RMSE (mm)	0.366	0.372	0.608	**0.325**	0.330
Heading med. (°)	0.51	0.47	**0.41**	1.36	1.10
Jerk RMS (rad s^{-3})	757	773	745	**390**	521
Samples settling (%)	51	64	48	18	**17**
Med. speed (mm s^{-1})	6.9	7.5	7.8	12.4	**14.1**
Sim. draw time (s)	76.8	**70.1**	101.0	73.4	71.3
Latency p50 (ms)	-	-	-	**9.1**	17.7

TABLE III

GOLD TEST SET PER CLASS: MEAN PATH RMSE IN MM (FINISHED / TOTAL WHERE NOT ALL FINISHED).

Class	n	Lula IK	S-IK	ACT	SmolVLA
Bird	9	0.266	0.260	0.191	**0.184**
Cat	12	0.275	0.267	0.195	**0.188**
Dog	13	0.282 (12)	0.270	0.189	**0.176**
Fish	4	0.295 (3)	0.269	0.198	**0.161**
Frog	5	0.289 (4)	0.271	0.183	**0.169**
Horse	5	0.270	0.266	0.198	**0.188**

ACT's progress rule with no settle gate. It finishes all 48 drawings (IK's timeouts came from the gate), but its path RMSE improves on IK by only 0.010 mm (0.267 mm, lower on 37 of 45), its corner error grows from 0.366 to 0.608 mm because pure pursuit cuts corners, and its jerk stays at IK's level. ACT is lower than streaming IK on all 48 drawings (paired reduction 0.075 mm, CI 0.069 to 0.080). Slowing streaming IK to IK's own pace (carrot limit 6.9 mm s^{-1}) does not help: 0.266 mm RMSE, jerk down to 524 but corner error up to 0.973 mm, and 181 s per drawing, still above ACT on all 48 drawings. Removing the gate is not what makes ACT accurate; the learned policy is. Nor is gravity sag: adding the servo's steady-state sag, $-\tau_g(q)/k_p$ per joint, to IK's targets lowers its RMSE on all 34 drawings both versions finish, but only by 0.011 mm (0.266 mm over 37 finished), and it times out on 11 of 48, with the pen inside the position tolerance but moving just above the 7 mm s^{-1} settle threshold; ACT is lower on all 37 (paired reduction 0.074 mm, $p = 1.5 \times 10^{-11}$). Streaming IK's longer draw time comes from its speed cap, which also slows the pen-up moves (46.9 against 21.2 s per drawing); its pen-down time is shorter than IK's.

Noise injection is essential. We retrained ACT on the same 8,490 episodes with the DART noise and joint offset removed (identical drawings, height jitter, architecture, seed and 40k steps). Its training loss is lower (0.21 against 0.42), but in closed loop it drifts past the 5 mm safety limit within the first stroke on all 48 drawings (median 36 s simulated): without off-path states in training, the first small error compounds, the covariate shift DART was designed for [21].

TABLE IV

DEVELOPMENT HISTORY OF THE ACT PATH FOLLOWER (48 GOLD DRAWINGS, 1 G, MEANS OVER FINISHED DRAWINGS). EACH ROW WAS RUN ONCE DURING DEVELOPMENT WITH THE CODE OF ITS TIME, SO ROWS ARE NOT A CONTROLLED ABLATION. ROWS ABOVE THE RULE USE THE DEFAULT SIM SCENE, ROWS BELOW THE STIFFENED LAB-ARM TWIN.

Controller	Fin.	RMSE	Max	Jerk
Lula IK	39/48	0.698	0.985	806
ACT, 30 Hz, full step held 2 ticks	48/48	0.366	2.599	1018
+ 60 Hz queries, half step (Eq. 3)	48/48	0.325	2.360	272
+ progress capped at 5 mm (hairpins)	48/48	**0.239**	1.472	**255**
+ cap at 1.5 mm (spike tips)	48/48	0.256	**0.885**	285
Lula IK, lab-arm twin	44/48	0.283	**0.775**	752
ACT, lab-arm twin, height jitter	**48/48**	**0.195**	0.812	**390**

What a viewer sees. A 0.08 mm RMSE gain is about a sixth of a 0.5 mm marker line, so we also measure the line's shape from the signed lateral deviation of the pen from the path. ACT sits closer to the path on average (lateral RMS 114 against 141 μm, closer on 44 of 45 drawings), but it wobbles more at fine scale: after removing drift longer than 3 mm, its residual is 78 against 52 μm, with 23 against 12 reversals larger than 0.25 mm per 100 mm of ink, worse on all 45 drawings. Both stay well inside the line: only 0.12% (ACT) and 0.04% (IK) of the path is more than a line width off, and SmolVLA wobbles little (56 μm, 0.005%). The same wobble explains IK's lower heading error (median 0.51° against 1.36°): filtering out wobble shorter than 3 mm lowers ACT's to 0.77°, the error does not depend on speed (1.3 to 1.5° in every speed bin), and IK's median is pulled down by the 18% of its samples spent nearly stationary at a waypoint (1.2% for ACT). ACT therefore lowers mean tracking error and joint jerk, but does not draw a smoother line at sub-millimetre scale.

What mattered. Table IV traces the design. Querying at 60 Hz with half steps cut jerk 3.7×; capping the progress projection fixed hairpin jumps (mean worst point 2.36 to 1.47 mm), and a tighter cap fixed spike tips (0.885 mm) at a small RMSE cost. Moving to the stiffened lab-arm twin, with paper-height jitter in training, gives the final policy. Each variant beats IK on path RMSE in its own scene.

Paper placement. Table V moves the sheet on a 16-drawing subset. ACT, trained with height jitter, finishes all 64 runs within 0.189 to 0.192 mm, and SmolVLA, trained on the same jittered data, stays within 0.171 to 0.182 mm. IK drops to 12/16 with the paper 12 mm lower and degrades to 0.299 mm under a planar shift; streaming IK finishes every run but degrades the same way under the shift (0.261 to 0.286 mm).

SmolVLA against ACT. On the full gold set with the batch runner (Table II), SmolVLA stage 2, served with zero

TABLE V

PAPER PLACEMENT ROBUSTNESS, 16 GOLD DRAWINGS: MEAN PATH RMSE IN MM (FINISHED DRAWINGS IN BRACKETS WHERE FEWER THAN 16), FINAL CODE FOR ALL FOUR CONTROLLERS. S-IK: STREAMING IK.

Placement	Lula IK	S-IK	ACT	SmolVLA
Nominal	0.269 (15)	0.261	0.191	0.178
12 mm lower	0.279 (12)	0.265	0.189	0.171
12 mm higher	0.267 (15)	0.255	0.192	0.182
Shifted $(+15, -15)$ mm	0.299 (15)	0.286	0.189	0.182

TABLE VI

SMOLVLA AND ACT, INTERACTIVE RUNNER. MEAN PATH RMSE / P95 (MM); ALL RUNS FINISHED.

Controller	6 animals	6 faces
SmolVLA, 3 actions per chunk	0.327 / 0.581	-
SmolVLA, re-plan every tick	0.230 / 0.410	0.288 / 0.473
ACT	**0.206 / 0.346**	**0.252 / 0.411**
Hybrid ACT, no skips	0.210 / 0.336	0.253 / 0.410
Hybrid ACT, injected skips	0.212 / 0.344	0.256 / 0.421
Hybrid SmolVLA, injected skips	0.244 / 0.409	-

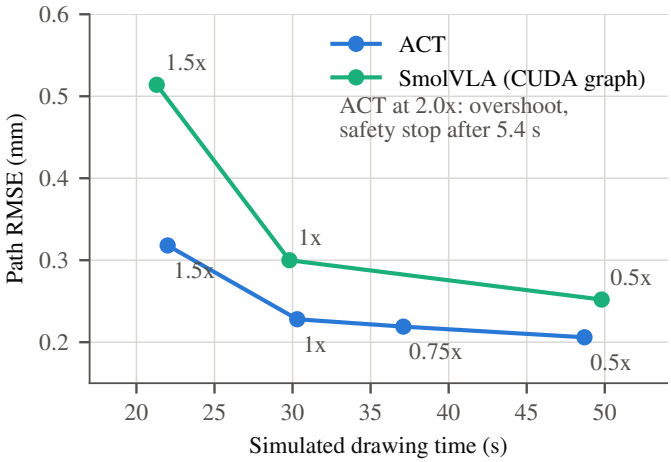


Fig. 7. Speed against accuracy on the gold cat (interactive runner): the step scale α of Eq. (3) relative to the default 0.5 is shown next to each point. ACT keeps most of its accuracy up to 1.0; SmolVLA degrades sooner.

noise, three flow steps and a CUDA graph and re-planned every tick, is as accurate as ACT: 0.180 against 0.192 mm, lower on 36 of 48 drawings (paired reduction 0.012 mm, CI 0.008 to 0.017, $p = 5 \times 10^{-6}$), but within ACT's seed spread ($\pm$0.013 mm), and by segment error only 0.007 mm lower (32 of 48). It pays for it with a third more joint jerk (521 against 390), a higher worst point (0.871 against 0.776 mm), and twice the query latency (17.7 against 9.1 ms). In the interactive runner used for the studies below the order was reversed (0.230 against 0.206 mm on six animals, 10 flow steps, and 0.252 against 0.206 mm on one cat with the same three-step CUDA-graph server). The runners share the policy servers but have separate control loops. The interactive runner lacks the batch runner's 7 mm s^{-1} speed check before advancing the target, and draws about 1.5$\times$ faster. Removing that check in the batch runner on the same six animals reproduces the faster pace (71 to 52 s) and the higher error for both policies (ACT 0.193 to 0.258 mm, SmolVLA 0.178 to 0.248 mm), but SmolVLA still leads on five of six, also by a segment-based error that removes sampling density. The speed check explains the interactive runner's pace and error, not the ranking; we have not isolated what flips it, so we compare only within a runner. The interactive-runner studies use one gold drawing per class and six face sketches unseen by both policies (Table VI). Offline, on 1,024 held-out frames, ACT's first-action error is 0.28 (normalised so that predicting zero scores 1.0) against 0.36 for SmolVLA. In closed loop (10 flow steps), executing three actions per chunk gives 0.327 mm; re-planning every tick with the same checkpoint gives 0.230 mm, within 12% of ACT. Most of the gap was therefore execution, not prediction. On faces, SmolVLA is 14% behind ACT. Both generalise from animals to faces at about +22 to +25% RMSE. Faces were never in either policy's training data. In the batch runner, on nine face sketches (15 to 20 strokes), ACT draws at 0.212 mm against IK's 0.302 mm, lower on all nine, with half the joint jerk (456 against 964), about 10% above its error on the animal test set.

End-to-end use. The pipeline and controllers are wrapped in WildTrace Studio, a web application in which a user asks for an animal in chat or uploads a photo, picks a face, and watches the twin draw it live with camera feeds and a live pen trace. On a 21-stroke portrait started this way, ACT drew at 0.241 mm in 143 s of simulated time against IK's 0.408 mm in 286 s; reaching nearby strokes at pen-up height rather than full approach height then cut ACT's time by 27% with unchanged error.

Camera-conditioned SmolVLA (negative result). Replacing the rendered path with synthetic overhead and tip-crop camera views improved offline error (0.306) but not closed-loop tracking, and the policy ignored the page: blanking it changed first-action error by less than 0.001. Gap-labelled data, two DAgger rounds [22], retrace weighting and an explicit redraw mode taught the policy to reverse direction at a gap (2% of its going-back actions pointed the wrong way, against 38% for ACT), but not to land back on the line: the camera policies abandoned 16 of 17 self-directed redraws.

Hybrid repair. On all 48 gold drawings in the batch runner, the supervisor never intervened without injected faults, and ACT's results were unchanged. With the pen made to skip ink on purpose (3 to 8 mm skips every 60 mm of inked travel on average, 345 skips in all), it made 232 redraws, gave up none, and left 0.24% of the path uninked on average; the remaining skips fell on line already re-inked or were shorter than the gap threshold. The cost was 0.001 mm of path RMSE and about 1 s per drawing, with joint jerk rising from 390 to 510 during IK's redraw moves.

Speed. Scaling ACT's applied step (Fig. 7) from 0.5 to 0.75 of the prediction keeps RMSE within 6% (0.206 to 0.219 mm) and cuts simulated time by 24%; 1.0 gives +11% error for 38% less time; 2.0 overshoots and trips the 5 mm safety

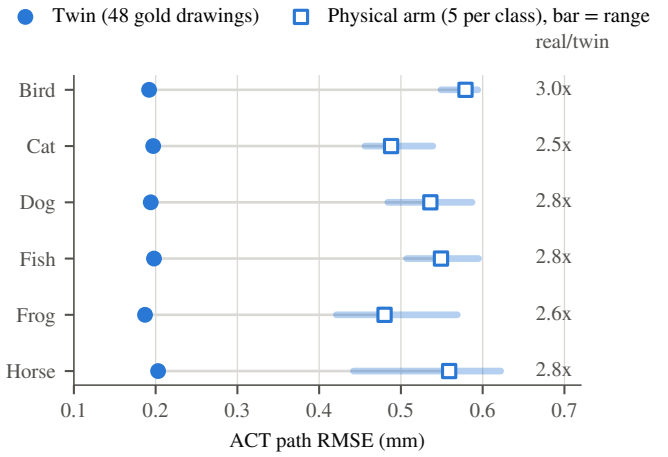


Fig. 8. ACT path RMSE per class in the twin (gold set) and on the physical arm (five drawings per class, range shown). The real arm is 2.5 to 3.0× the twin in every class. Drawings differ between the two sets, so the comparison is per class, not paired.

stop, consistent with the curvature-speed trade-off reported for Piktor-O-Bot [13]. SmolVLA has no faster setting within 10%.

Physical arm. The twin-trained ACT policy ran on the lab arm without fine-tuning. In a 30-drawing batch (five per class, not the gold-set drawings) every drawing finished, at 0.532 mm mean path RMSE (0.421 to 0.622 mm), 2.7× the twin (Table VII, Fig. 8). Stalls were the main failure mode: 14 of 30 drawings needed at least one nudge to resume (26 in total), always mid-stroke where waypoints are dense; the pen sat about 0.5 mm beside the path and corrected only sideways. Three runs of two face portraits drawn with markers (Table VIII) finished at 0.497 mm measured RMSE, using an automatic 0.5 s stall release instead of nudges.

Scoring the ink. The arm's reported tip says where the arm believes the pen is, not what reached the paper. We therefore score the sheet itself: the page is found in an overhead photograph and rectified to millimetres, the ink is thinned to a centre line, and the ink is fitted to the plan by a trimmed rigid transform (ink onto plan, so missing ink cannot drag the fit). On portrait A the ink lies 0.43 to 0.48 mm from the plan on average, in line with the reported tip error, and 90% of the plan has ink within 2 mm. Portrait B scores lower for a reason the arm's log cannot see: the long hair below the shoulders inked faintly, so only 71% of the plan is covered. The photograph's 0.6 mm pixels bound this measurement; flatbed scans are planned for the paired real-arm study.

VI. Discussion and Limitations

Why ACT beats its own teacher. The ablations rule out two easy explanations. In the stiff lab-arm twin, gravity sag explains only 0.011 mm of IK's error, and removing the settle gate only 0.010 mm; most of IK's error is lag behind a slew-limited target that moves from waypoint to waypoint. ACT instead predicts joint increments from the next 2 cm of path, so it can lead the servo rather than chase it, and its noise-injected episodes teach it to steer back to the path: without them it fails

TABLE VII
ACT on the physical arm, 30 animal drawings (five per class). Path RMSE in mm.

Class	RMSE mean (range)	Worst point	Time (s)	Nudges / drawing
Bird	0.579 (0.549-0.594)	1.54	76	0.8
Cat	0.488 (0.456-0.539)	1.53	96	1.6
Dog	0.536 (0.484-0.587)	1.47	72	0.0
Fish	0.549 (0.507-0.595)	1.49	102	0.4
Frog	0.480 (0.421-0.569)	1.51	100	0.4
Horse	0.559 (0.442-0.622)	1.54	117	2.0
All	0.532 (0.421-0.622)	1.54	94	0.87

TABLE VIII
Face portraits on the physical arm with ACT and a marker. RMSE, p95 and max are measured from the arm's reported tip at 10 Hz; chamfer and coverage are measured from the ink in an overhead photograph. mm.

Run	RMSE	p95	Max	Chamfer	Cov. 2 mm	min
A (48 strokes), used sheet	0.486	0.912	1.306	0.74	90%	8.5
A (48 strokes), clean sheet	0.514	0.966	1.305	0.67	90%	8.3
B (59 strokes), clean sheet	0.490	0.905	1.273	2.13	71%	9.7

every drawing. Querying a short path window at 60 Hz also replaces IK's stop-and-settle with continuous motion, which is where IK's joint jerk comes from; the price is a fine lateral wobble that the 60 Hz half-step update does not damp. The streaming IK result shows that continuous motion alone is not enough: an IK solver chasing a moving target cuts corners and stays jerky, while the policy has learned from noise-injected labels how to correct toward the path.

Sim-to-real gap. The 2.7× gap is consistent across classes, which points to a systematic cause rather than content. Candidates we observed are a non-level sheet (line weight varies across the page), pen compliance, the arm's own position controller differing from the twin's servo, and dense-waypoint stalls. Fitting the paper plane from a three-point touch and training on stalled states are the next steps.

What the pipeline does not preserve. Recognisability and geometric fidelity are separate goals. For completed paper-r3 final images, mean silhouette IoU with the source photo is 0.560 for FLUX ($n = 20$) and 0.225 for OmniGen2 ($n = 20$, including 9 blank images); FLUX's facing direction matches in 55% of available cases. The redraw can preserve the animal category while changing its pose. Lower diffusion strength or stronger contour conditioning requires a separate evaluation.

Baseline strength. We compare against the waypoint-gated IK controller the system shipped with, a tuned version of it, a streaming IK that shares ACT's progress rule, and IK with gravity compensation; an optimisation-based tracker (for example model-predictive contouring control) and the vendor's own Cartesian motion commands on the real arm are stronger baselines we have not yet run. We also did not train a diffusion policy [4]: path following from a full path window has one correct action per state, the multimodality diffusion policy is built for is absent, and its iterative sampling adds the latency cost we measured for SmolVLA's flow head.

Limitations. Each controller was run once per drawing, which suffices in the twin: rerunning IK and ACT on six drawings reproduced every metric bit for bit; ACT was trained with three seeds; the other policies with one. The real-arm batch used different drawings from the gold set and relied on nudges, so it is not a controlled paired comparison. The path metric is geometric; ink scoring from photographs covers only three portrait runs so far, and perceived quality (a user study) is future work. IK was tuned only over the settings above. Real-arm portraits take 8 to 10 min, against 43 s to 2 min for systems built for exhibition speed [9]–[11]; our runs used the default step scale and stall releases, not a speed-tuned setting. The camera policies were evaluated on 12 drawings in the interactive runner only.

Reproducibility and ethics. Every number comes from per-run metric files that record configuration, checkpoint and runner; one script regenerates all figures and statistics. Code, configurations and metric files will be released under the MIT license (link withheld for review). Face photographs are CC BY 2.0 Open Images photographs credited in Fig. 2, and uploaded portraits are processed only locally.

VII. Conclusion

WildTrace turns photographs into robot pen drawings end to end. A compact ACT path follower trained only on kinematic IK labels draws held-out content more accurately than the IK solver that labelled it, even after tuning, with half its joint jerk, is robust to paper placement, and transfers to a physical xArm 7 at about half a millimetre RMSE. A streaming IK with the same progress rule gains almost nothing, so the accuracy comes from learning. A fine-tuned VLA re-planned every tick is as accurate but jerkier and slower to query, and a simple supervisor makes either policy robust to missed ink. Closing the remaining sim-to-real gap is our next goal.

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